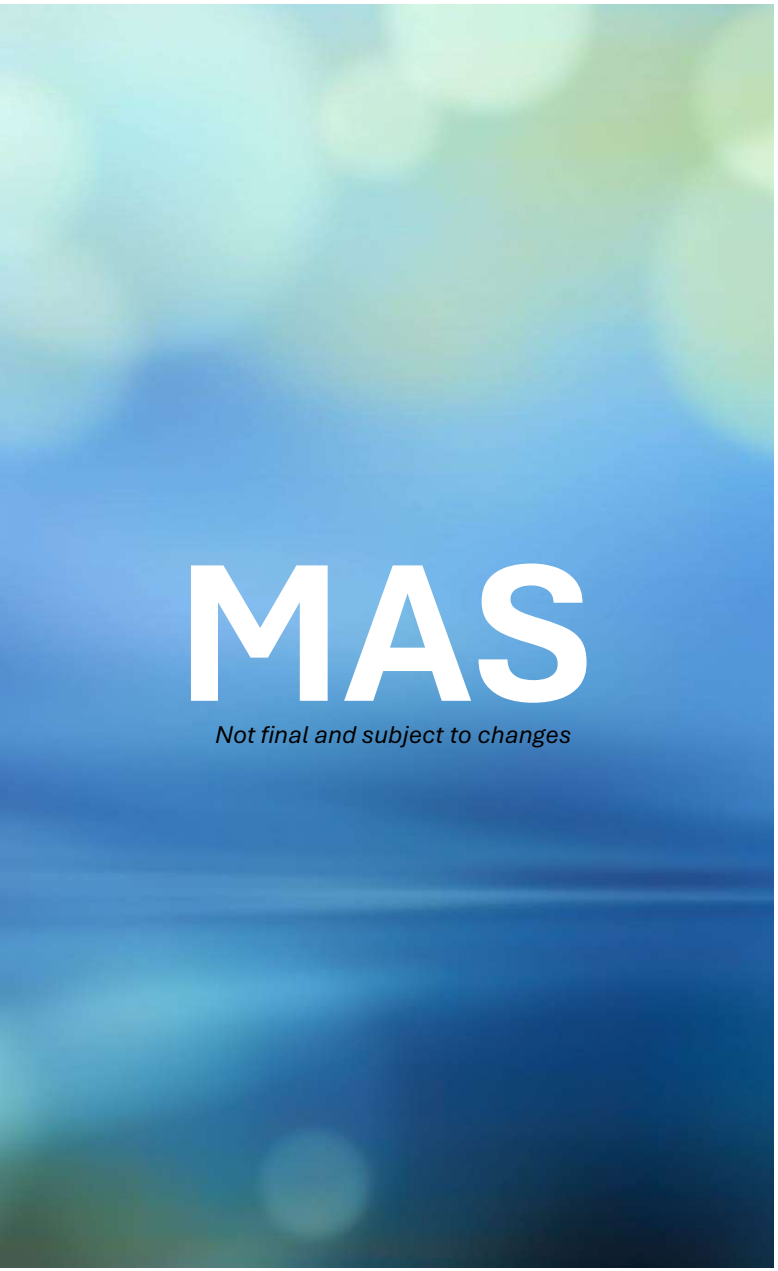
Mechanical **A**erial **S**ystems
(MAS)

F26, SDU Dronecenter

Robert Ladig – rola@sdu.dk



Not final and subject to changes

Unit 1 – Course goal, Intro, Housekeeping, Aerial Manipulation Intro, Simulator Setup, ROS2 (refresher ?!)
Unit 2 - Manipulator types, FK (refresher), IK (refresher), URDF (refresher?!), Simulator work

Unit 3 – Jacobians and coupled kinematics, Simulator work
Unit 4 – PID control (refresher!), Simulator implementation of coupled kinematics for SDU UAS system
Unit 5 – Material handover, Material check and prototyping of own aerial manipulator.

Unit 6 - Group presentation of concept, CAD and refined Unit 4 simulator

----------Flight lab is available to use and drones are RTF----------

Unit 7 – Intro to Flightlab, EIT playground, Offboard control test @ Flightlab
Unit 8 - Bench prototype of Aerial Manipulator
Unit 9 - Airframe integration
Unit 10 – Airframe integration, Flight test I, (opt.) Sensing and control integration

Unit 11 – Final group presentation, Flight test II
Unit 12 – Flight demo , Project report due

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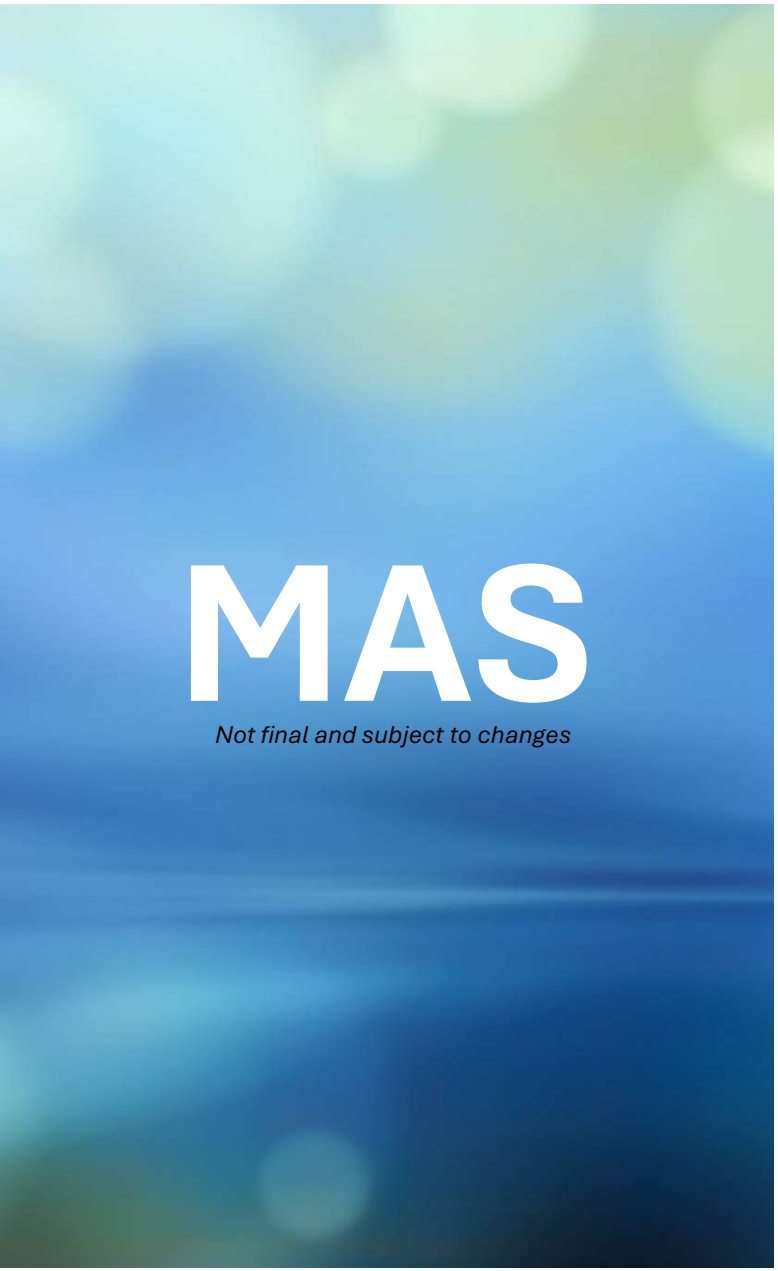

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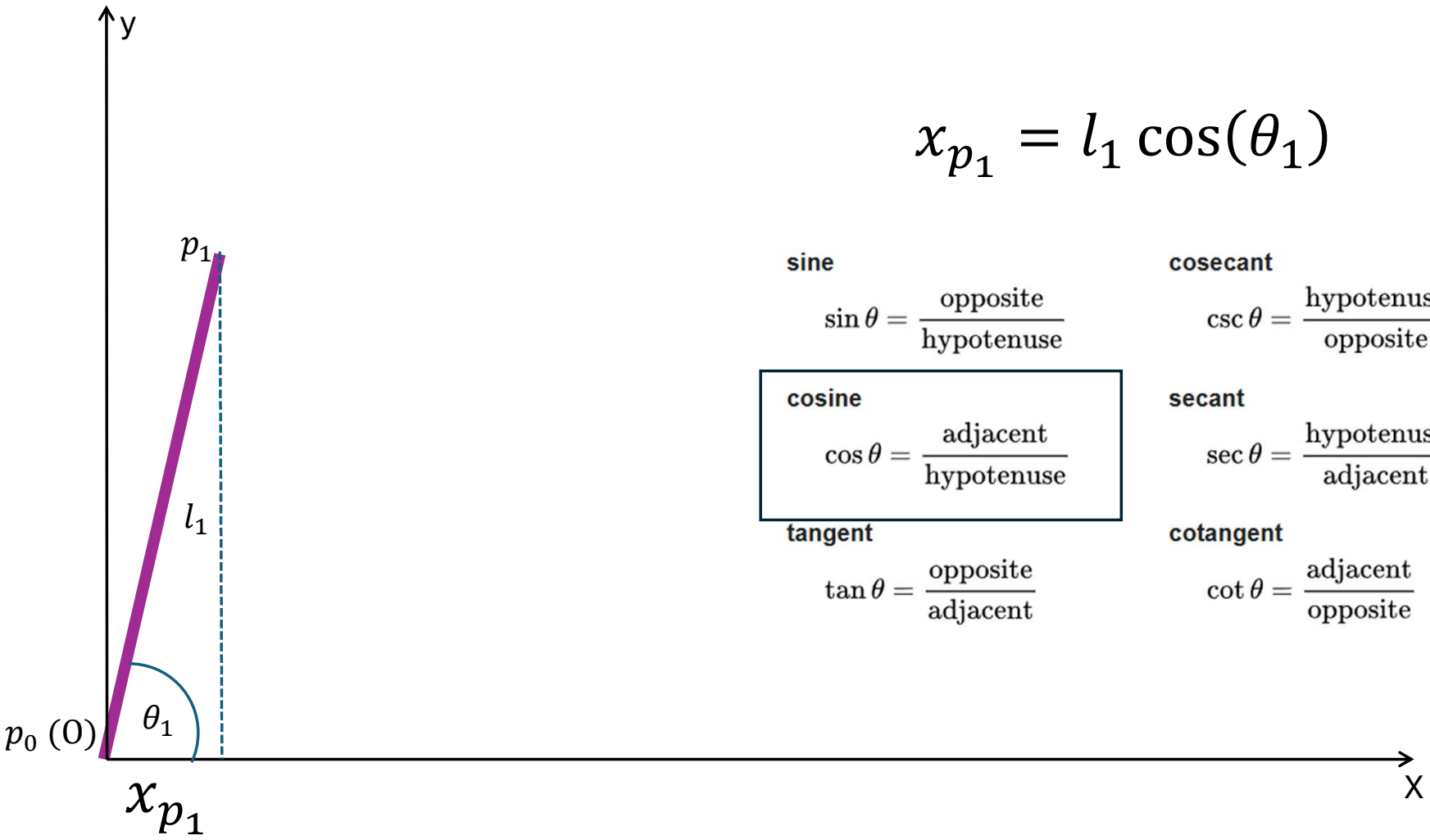


$$x_{p_1} = l_1 \cos(\theta_1)$$

sine

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

cosecant

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}$$

cosine

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

secant

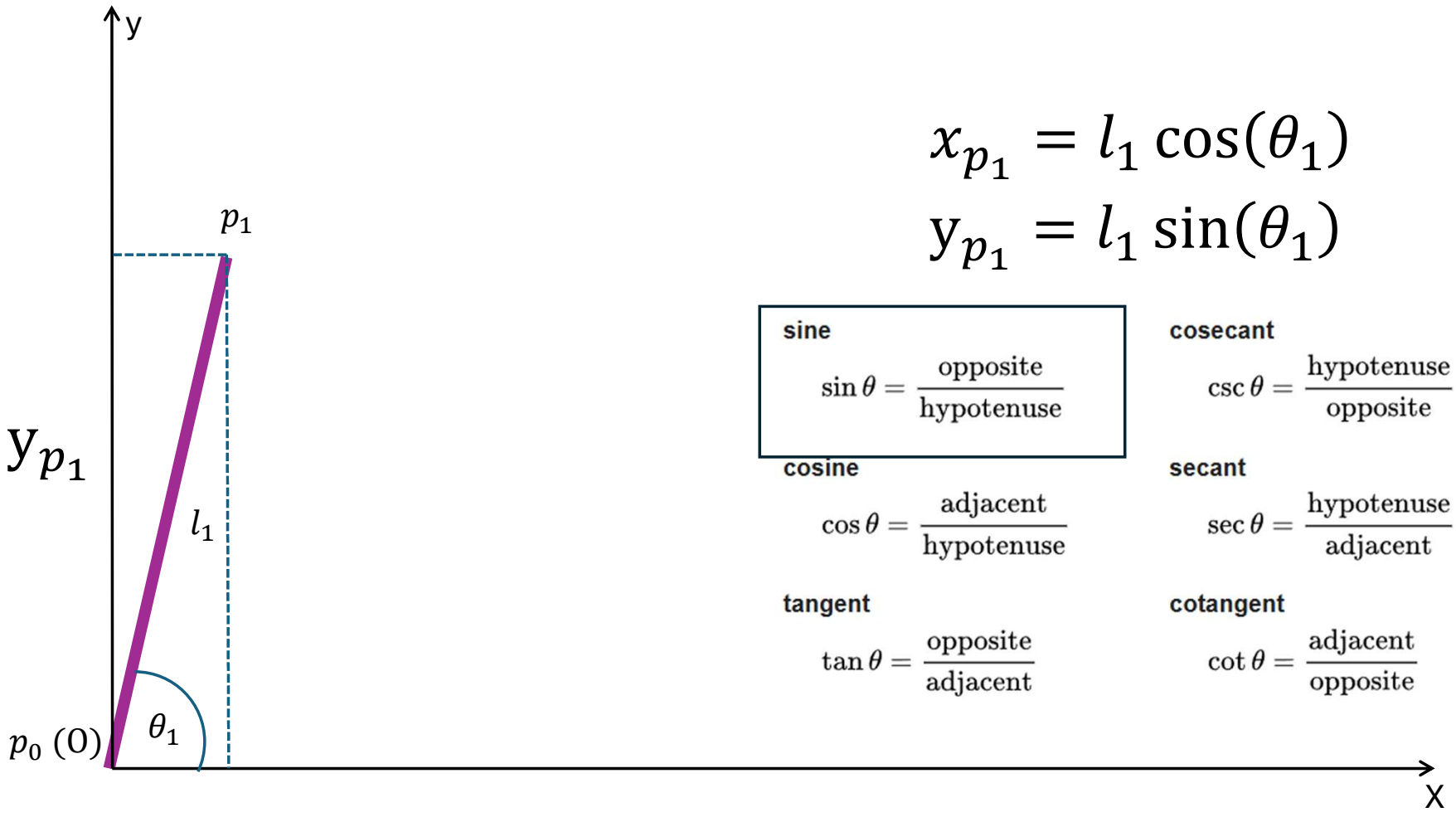
$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

tangent

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

cotangent

$$\cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$



$$x_{p_1} = l_1 \cos(\theta_1)$$
$$y_{p_1} = l_1 \sin(\theta_1)$$

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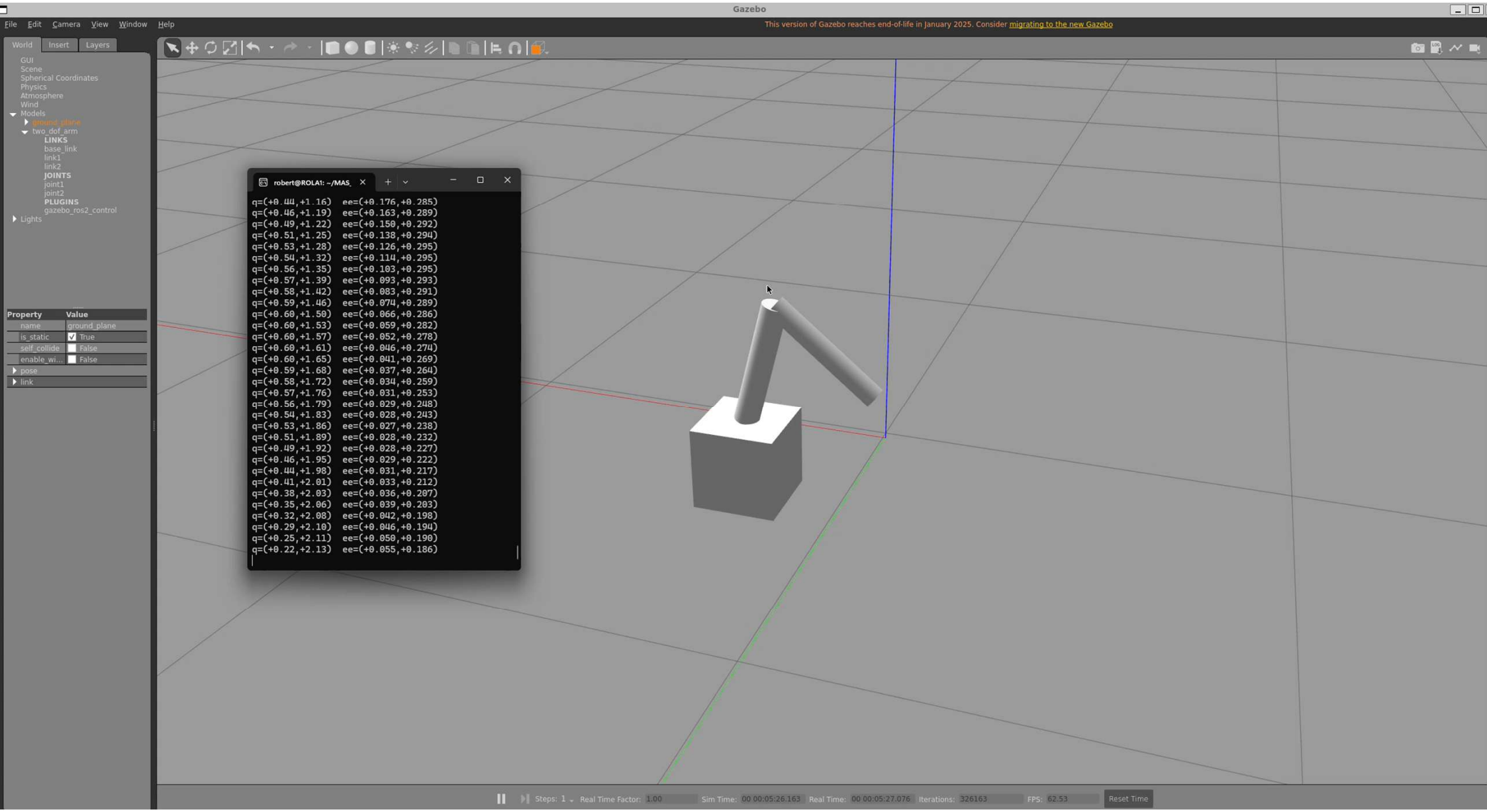
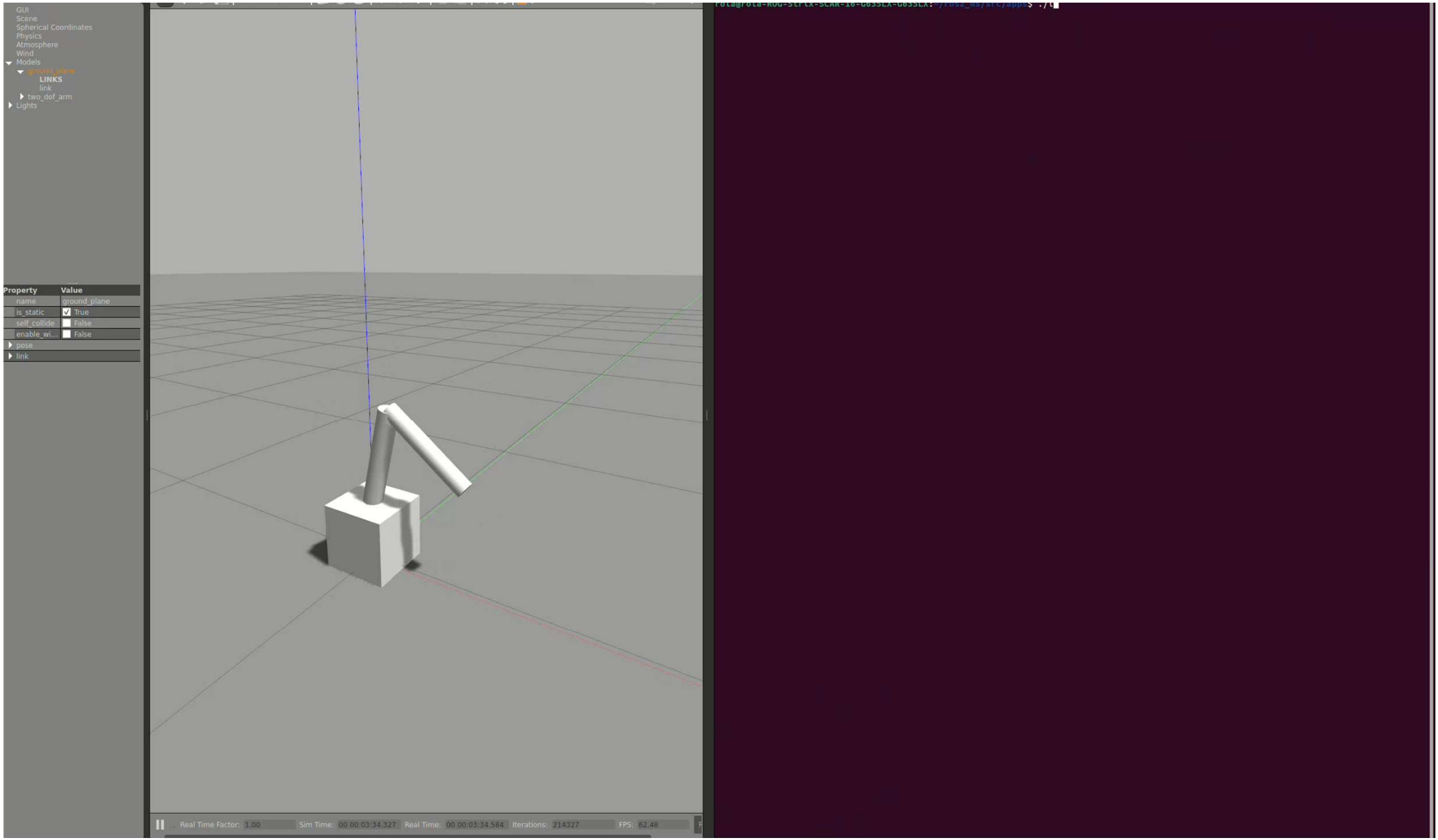
Forward Kinematics

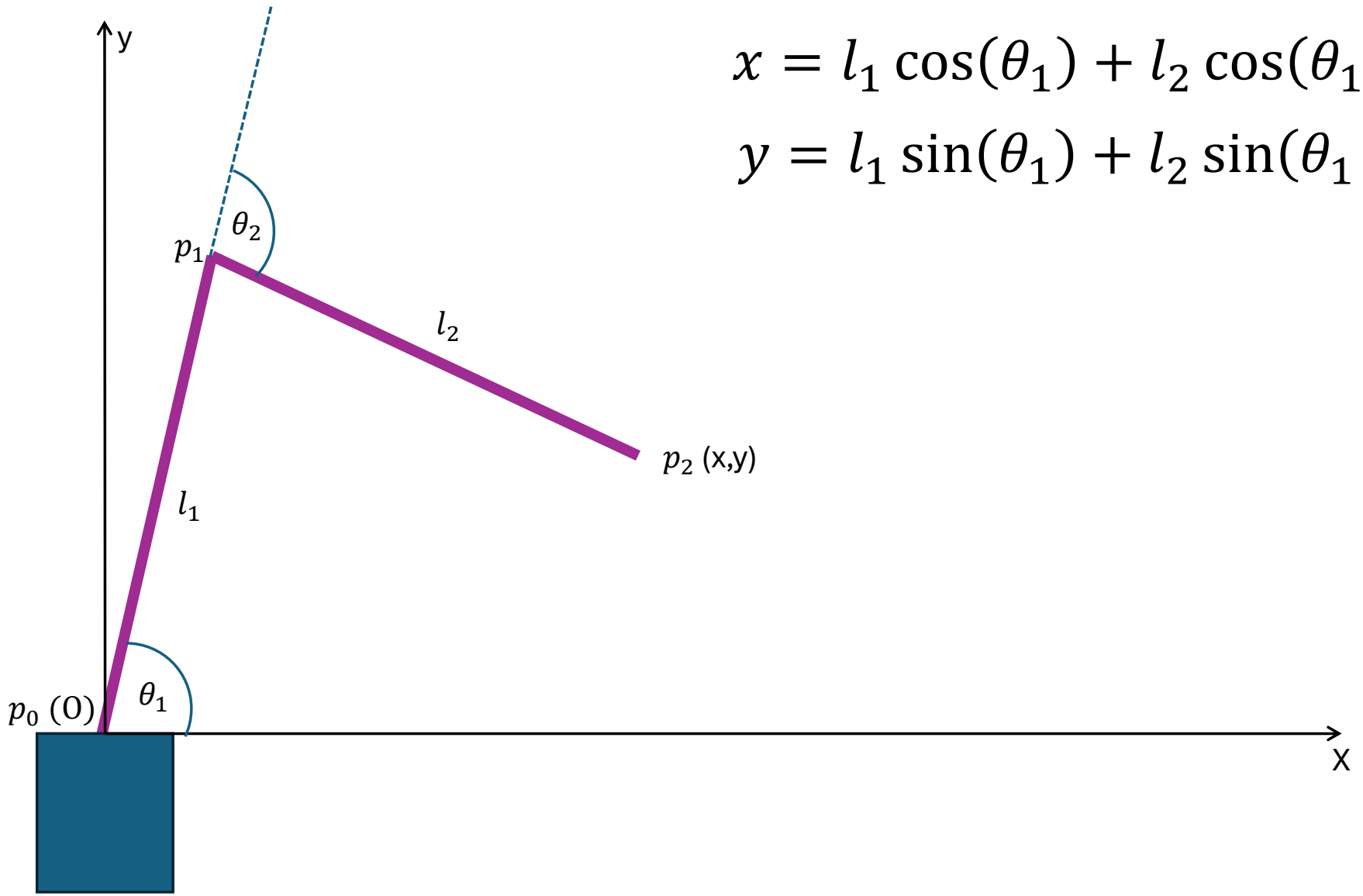
- “Where are we when we move our actuators?”


Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
two_dof_arm
LINKS
base_link
link1
link2
JOINTS
joint1
joint2
PLUGINS
gazebo_ros2_control
Lights
Property Value
name ground_plane
is_static True
self_collide False
enable_wi... False
pose
link
robert@ROLA1: ~/MAS_
q=(+0.44,+1.16) ee=(+0.176,+0.285)
q=(+0.46,+1.19) ee=(+0.163,+0.289)
q=(+0.49,+1.22) ee=(+0.150,+0.292)
q=(+0.51,+1.25) ee=(+0.138,+0.294)
q=(+0.53,+1.28) ee=(+0.126,+0.295)
q=(+0.54,+1.32) ee=(+0.114,+0.295)
q=(+0.56,+1.35) ee=(+0.103,+0.295)
q=(+0.57,+1.39) ee=(+0.093,+0.293)
q=(+0.58,+1.42) ee=(+0.083,+0.291)
q=(+0.59,+1.46) ee=(+0.074,+0.289)
q=(+0.60,+1.50) ee=(+0.066,+0.286)
q=(+0.60,+1.53) ee=(+0.059,+0.282)
q=(+0.60,+1.57) ee=(+0.052,+0.278)
q=(+0.60,+1.61) ee=(+0.046,+0.274)
q=(+0.60,+1.65) ee=(+0.041,+0.269)
q=(+0.59,+1.68) ee=(+0.037,+0.264)
q=(+0.58,+1.72) ee=(+0.034,+0.259)
q=(+0.57,+1.76) ee=(+0.031,+0.253)
q=(+0.56,+1.79) ee=(+0.029,+0.248)
q=(+0.54,+1.83) ee=(+0.028,+0.243)
q=(+0.53,+1.86) ee=(+0.027,+0.238)
q=(+0.51,+1.89) ee=(+0.028,+0.232)
q=(+0.49,+1.92) ee=(+0.028,+0.227)
q=(+0.46,+1.95) ee=(+0.029,+0.222)
q=(+0.44,+1.98) ee=(+0.031,+0.217)
q=(+0.41,+2.01) ee=(+0.033,+0.212)
q=(+0.38,+2.03) ee=(+0.036,+0.207)
q=(+0.35,+2.06) ee=(+0.039,+0.203)
q=(+0.32,+2.08) ee=(+0.042,+0.198)
q=(+0.29,+2.10) ee=(+0.046,+0.194)
q=(+0.25,+2.11) ee=(+0.050,+0.190)
q=(+0.22,+2.13) ee=(+0.055,+0.186)
Steps: 1 Real Time Factor: 1.00 Sim Time: 00 00:05:26.163 Real Time: 00 00:05:27.076 Iterations: 326163 FPS: 62.53 Reset Time


GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
ground_plane
LINKS
link
two_dof_arm
Lights
Property
Value
name
ground_plane
is_static
True
self_collide
False
enable_wi...
False
pose
link
Real Time Factor: 1.00
Sim Time: 00 00:03:34.327
Real Time: 00 00:03:34.584
Iterations: 214327
FPS: 62.48

Inverse Kinematics

- ~~“Where are we when we move our actuators?”~~
- “What positions have my actuators, when I am here?”



$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

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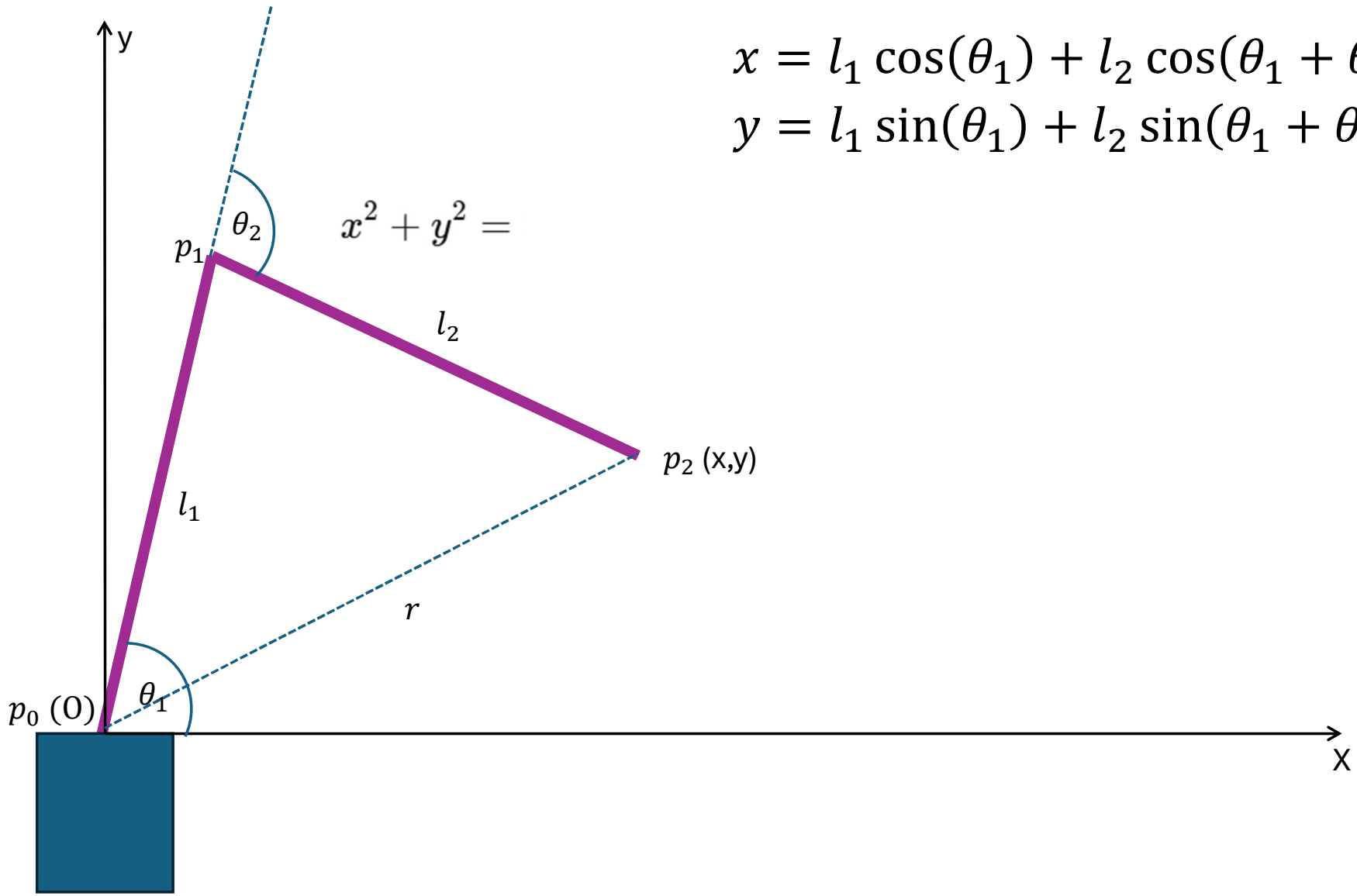
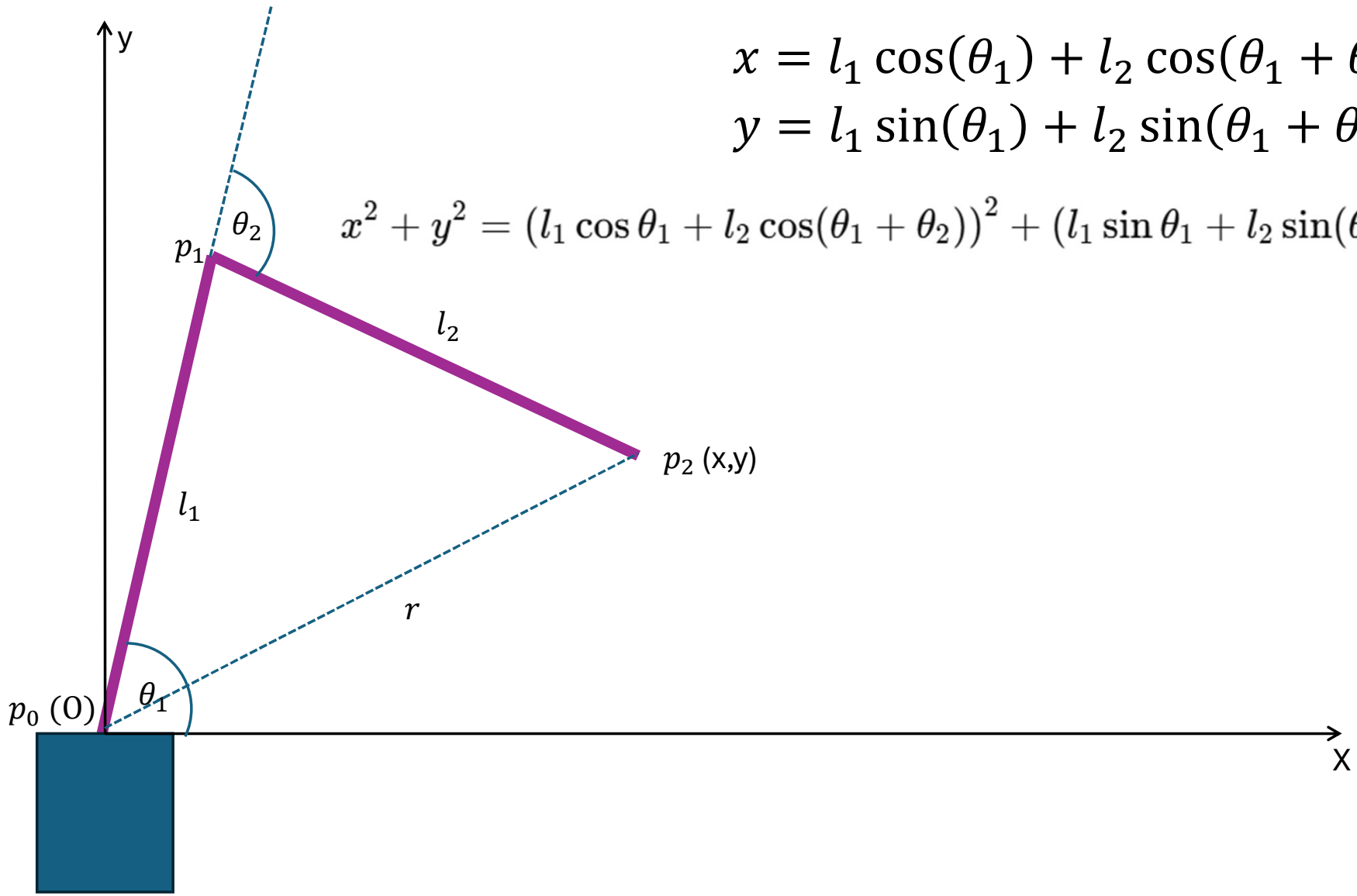


$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

$$x^2 + y^2 = (l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2))^2 + (l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2))^2$$



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$$x^2 + y^2 = l_1^2(\cos^2\theta_1 + \sin^2\theta_1) + l_2^2(\cos^2(\theta_1 + \theta_2) + \sin^2(\theta_1 + \theta_2)) + 2l_1l_2\big(\cos\theta_1 \cos(\theta_1 + \theta_2) + \sin\theta_1 \sin(\theta_1 + \theta_2)\big)$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
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$$\boxed{\begin{gathered}\cos^2 a + \sin^2 a = 1 \\ \cos a \cos b + \sin a \sin b = \cos(a - b)\end{gathered}}$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
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$$x^2 + y^2 = l_1^2\underbrace{(\cos^2\theta_1 + \sin^2\theta_1)}_{=1 \text{ (Identity 1)}} + l_2^2\underbrace{(\cos^2(\theta_1 + \theta_2) + \sin^2(\theta_1 + \theta_2))}_{=1 \text{ (Identity 1)}} + 2l_1l_2\underbrace{\big(\cos\theta_1 \cos(\theta_1 + \theta_2) + \sin\theta_1 \sin(\theta_1 + \theta_2)\big)}_{=\cos(\theta_1 - (\theta_1 + \theta_2)) \text{ (Identity 2)}}$$

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$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1l_2 \cos\big(\theta_1 - (\theta_1 + \theta_2)\big) =$$

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$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
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$$\boxed{x^2 + y^2 = l_1^2 + l_2^2 + 2l_1l_2 \cos\theta_2}$$

$$\boxed{x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 \cos\theta_2}$$

$$\boxed{x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 \cos\theta_2}$$

$$\cos\theta_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

$$\boxed{x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 \cos\theta_2}$$

$$\cos\theta_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1 l_2}$$

$$\boxed{\theta_2 = \operatorname{atan2}\left(\pm\sqrt{1 - \cos^2\theta_2},\ \cos\theta_2\right)}$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
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$$\boxed{\begin{aligned} \cos(\theta_1 + \theta_2) &= \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2 \\ \sin(\theta_1 + \theta_2) &= \sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2 \end{aligned}}$$

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$$x = l_1 \cos\theta_1 + l_2(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2)$$

$$x = (l_1 + l_2 \cos\theta_2) \cos\theta_1 - (l_2 \sin\theta_2) \sin\theta_1$$

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$$x = l_1 \cos\theta_1 + l_2(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2)$$

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$$y = l_1 \sin\theta_1 + l_2(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2)$$

$$y = (l_1 + l_2 \cos\theta_2) \sin\theta_1 + (l_2 \sin\theta_2) \cos\theta_1$$

$$\boxed{k_1 = l_1 + l_2 \cos\theta_2, \qquad k_2 = l_2 \sin\theta_2}$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
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$$\boxed{\begin{aligned} \cos(\theta_1 + \theta_2) &= \cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2 \\ \sin(\theta_1 + \theta_2) &= \sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2 \end{aligned}}$$

$$x = l_1 \cos\theta_1 + l_2(\cos\theta_1 \cos\theta_2 - \sin\theta_1 \sin\theta_2)$$

$$x = (l_1 + l_2 \cos\theta_2) \cos\theta_1 - (l_2 \sin\theta_2) \sin\theta_1$$

$$y = l_1 \sin\theta_1 + l_2(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2)$$

$$y = (l_1 + l_2 \cos\theta_2) \sin\theta_1 + (l_2 \sin\theta_2) \cos\theta_1$$

$$\boxed{k_1 = l_1 + l_2 \cos\theta_2, \qquad k_2 = l_2 \sin\theta_2}$$

$$x = k_1 \cos\theta_1 - k_2 \sin\theta_1$$

$$y = k_1 \sin\theta_1 + k_2 \cos\theta_1$$

$$\boxed{k_1 = l_1 + l_2 \cos\theta_2, \qquad k_2 = l_2 \sin\theta_2}$$

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$$x = k_1 \cos\theta_1 - k_2 \sin\theta_1$$

$$y = k_1 \sin\theta_1 + k_2 \cos\theta_1$$

$$\boxed{\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}}$$

$$\boxed{k_1 = l_1 + l_2 \cos\theta_2, \qquad k_2 = l_2 \sin\theta_2}$$

$$x = k_1 \cos\theta_1 - k_2 \sin\theta_1$$

$$y = k_1 \sin\theta_1 + k_2 \cos\theta_1$$

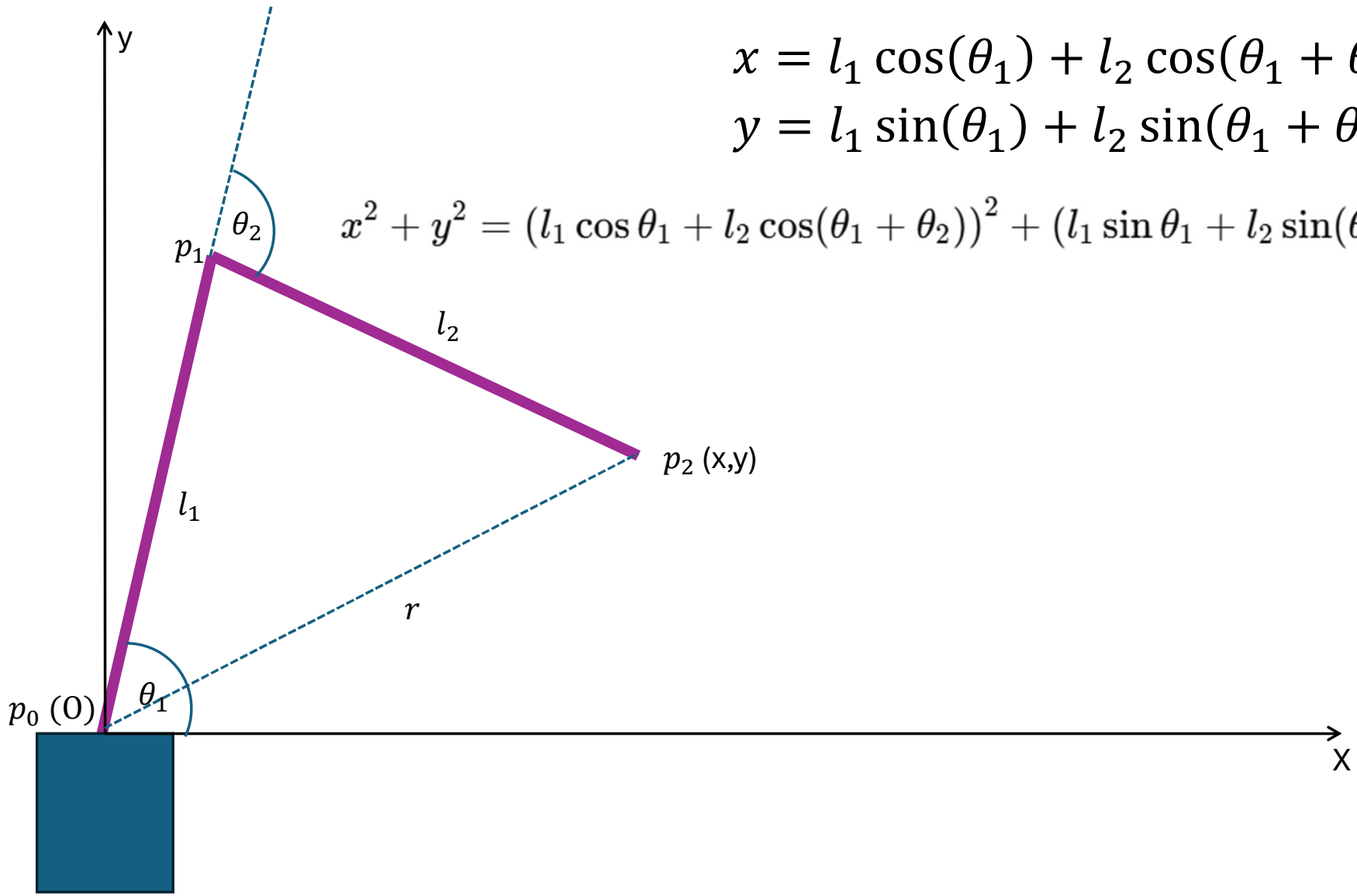
$$\boxed{\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}}$$

$$\mathrm{atan2}(y, x) = \theta_1 + \mathrm{atan2}(k_2, k_1)$$

$$\boxed{\theta_1 = \mathrm{atan2}(y, x) - \mathrm{atan2}(k_2, k_1)}$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

$$x^2 + y^2 = (l_1 \cos\theta_1 + l_2 \cos(\theta_1 + \theta_2))^2 + (l_1 \sin\theta_1 + l_2 \sin(\theta_1 + \theta_2))^2$$



$$x^2 + y^2 = l_1^2(\cos^2\theta_1 + \sin^2\theta_1) + l_2^2(\cos^2(\theta_1+\theta_2) + \sin^2(\theta_1+\theta_2)) + 2l_1l_2\big(\cos\theta_1\cos(\theta_1+\theta_2) + \sin\theta_1\sin(\theta_1+\theta_2)\big)$$

Now for 3 link arm: ...

$$x = l_1\cos\theta_1 + l_2\cos(\theta_1+\theta_2) + l_3\cos(\theta_1+\theta_2+\theta_3)$$

$$y = l_1\sin\theta_1 + l_2\sin(\theta_1+\theta_2) + l_3\sin(\theta_1+\theta_2+\theta_3)$$

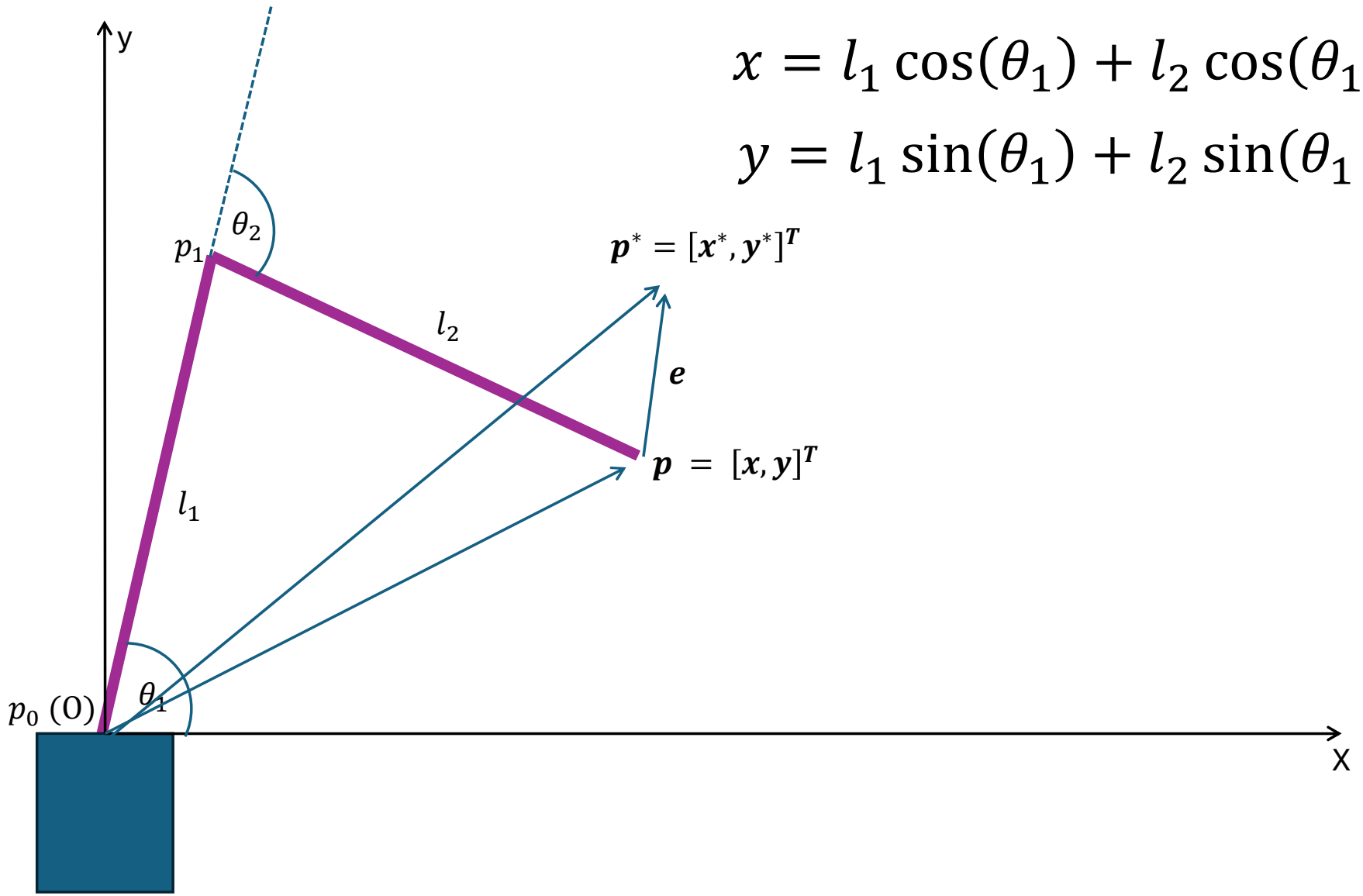
$$\begin{aligned} x^2 + y^2 = {} & l_1^2\big(\cos^2\theta_1 + \sin^2\theta_1\big) + l_2^2\big(\cos^2(\theta_1+\theta_2) + \sin^2(\theta_1+\theta_2)\big) + l_3^2\big(\cos^2(\theta_1+\theta_2+\theta_3) + \sin^2(\theta_1+\theta_2+\theta_3)\big) \\ & + 2l_1l_2\Big(\cos\theta_1\cos(\theta_1+\theta_2) + \sin\theta_1\sin(\theta_1+\theta_2)\Big) \\ & + 2l_1l_3\Big(\cos\theta_1\cos(\theta_1+\theta_2+\theta_3) + \sin\theta_1\sin(\theta_1+\theta_2+\theta_3)\Big) \\ & + 2l_2l_3\Big(\cos(\theta_1+\theta_2)\cos(\theta_1+\theta_2+\theta_3) + \sin(\theta_1+\theta_2)\sin(\theta_1+\theta_2+\theta_3)\Big) \end{aligned}$$

Jacobian

- ~~“Where are we when we move our actuators?”~~
- ~~“What positions have my actuators, when I am here?”~~
- “If I want to move my end effector an infinitesimally small amount in a certain direction, how should my actuators move?”

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

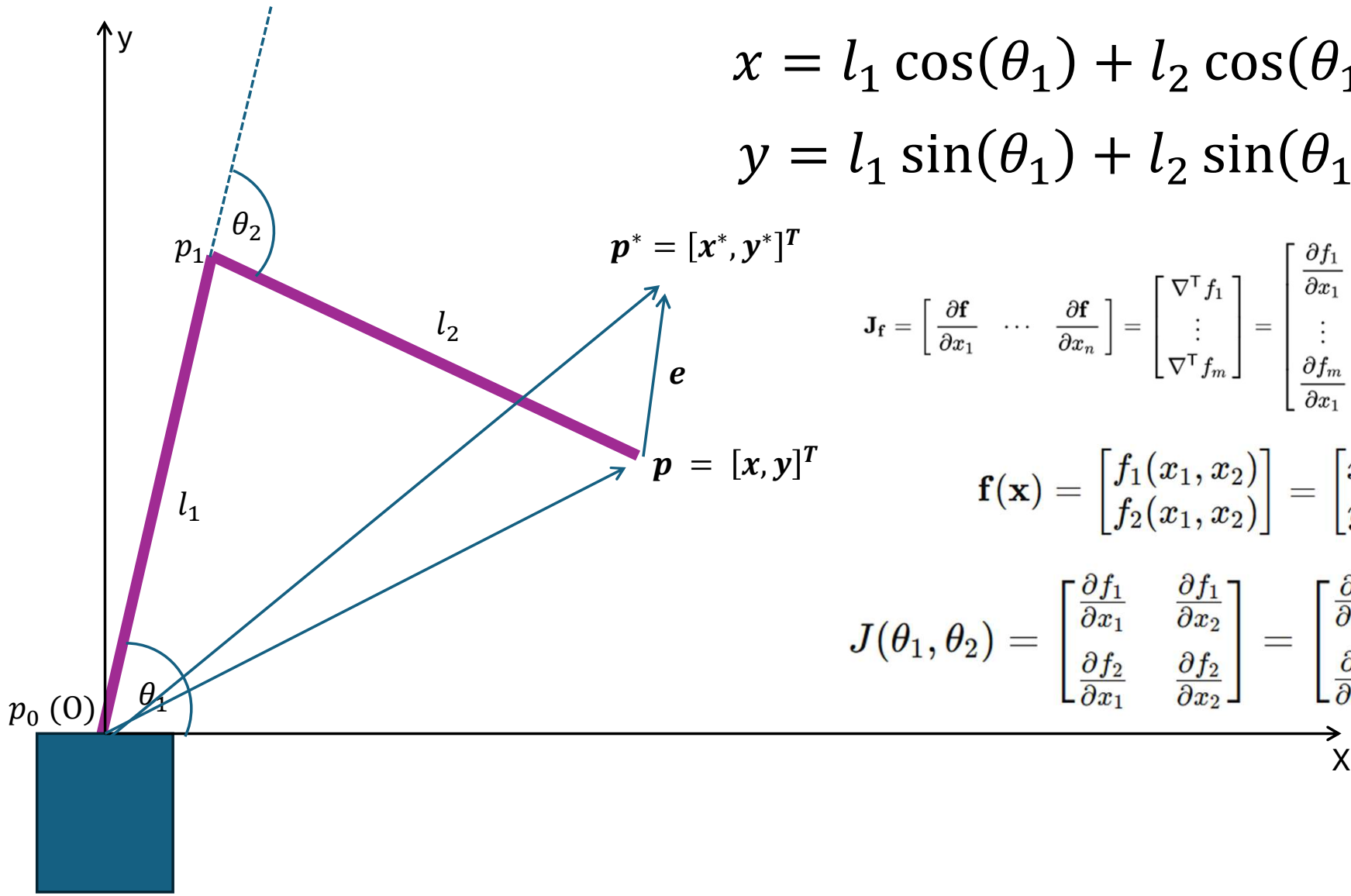
$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$



Definition [edit]

Let $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$ be a function such that each of its first-order partial derivatives exists on $\mathbb{R}^n$. This function takes a point $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$ as input and produces the vector $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})) \in \mathbb{R}^m$ as output. Then the Jacobian matrix of $\mathbf{f}$, denoted $\mathbf{J_f}$, is the $m \times n$ matrix whose (i, j) entry is $\frac{\partial f_i}{\partial x_j}$; explicitly

$$\mathbf{J_f} = \begin{bmatrix} \dfrac{\partial \mathbf{f}}{\partial x_1} & \cdots & \dfrac{\partial \mathbf{f}}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla^\mathsf{T} f_1 \\ \vdots \\ \nabla^\mathsf{T} f_m \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{bmatrix}$$



$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

$$\mathbf{J_f} = \begin{bmatrix} \frac{\partial \mathbf{f}}{\partial x_1} & \cdots & \frac{\partial \mathbf{f}}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla^{\mathsf{T}} f_1 \\ \vdots \\ \nabla^{\mathsf{T}} f_m \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(x_1, x_2) \\ f_2(x_1, x_2) \end{bmatrix} = \begin{bmatrix} x(\theta_1, \theta_2) \\ y(\theta_1, \theta_2) \end{bmatrix}$$

$$J(\theta_1, \theta_2) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$
$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

$$J(\theta_1, \theta_2) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$

$$\boxed{\frac{d}{dt}\cos(\theta) = -\sin(\theta)\,\dot{\theta} \qquad \frac{d}{dt}\sin(\theta) = \cos(\theta)\,\dot{\theta}}$$

$$J(\theta_1, \theta_2) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial \theta_2} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial \theta_2} \end{bmatrix}$$

$$x = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2)$$

$$\boxed{\frac{d}{dt}\cos(\theta) = -\sin(\theta)\,\dot{\theta} \qquad \frac{d}{dt}\sin(\theta) = \cos(\theta)\,\dot{\theta}}$$

$$\frac{\partial x}{\partial \theta_1} = -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2)$$

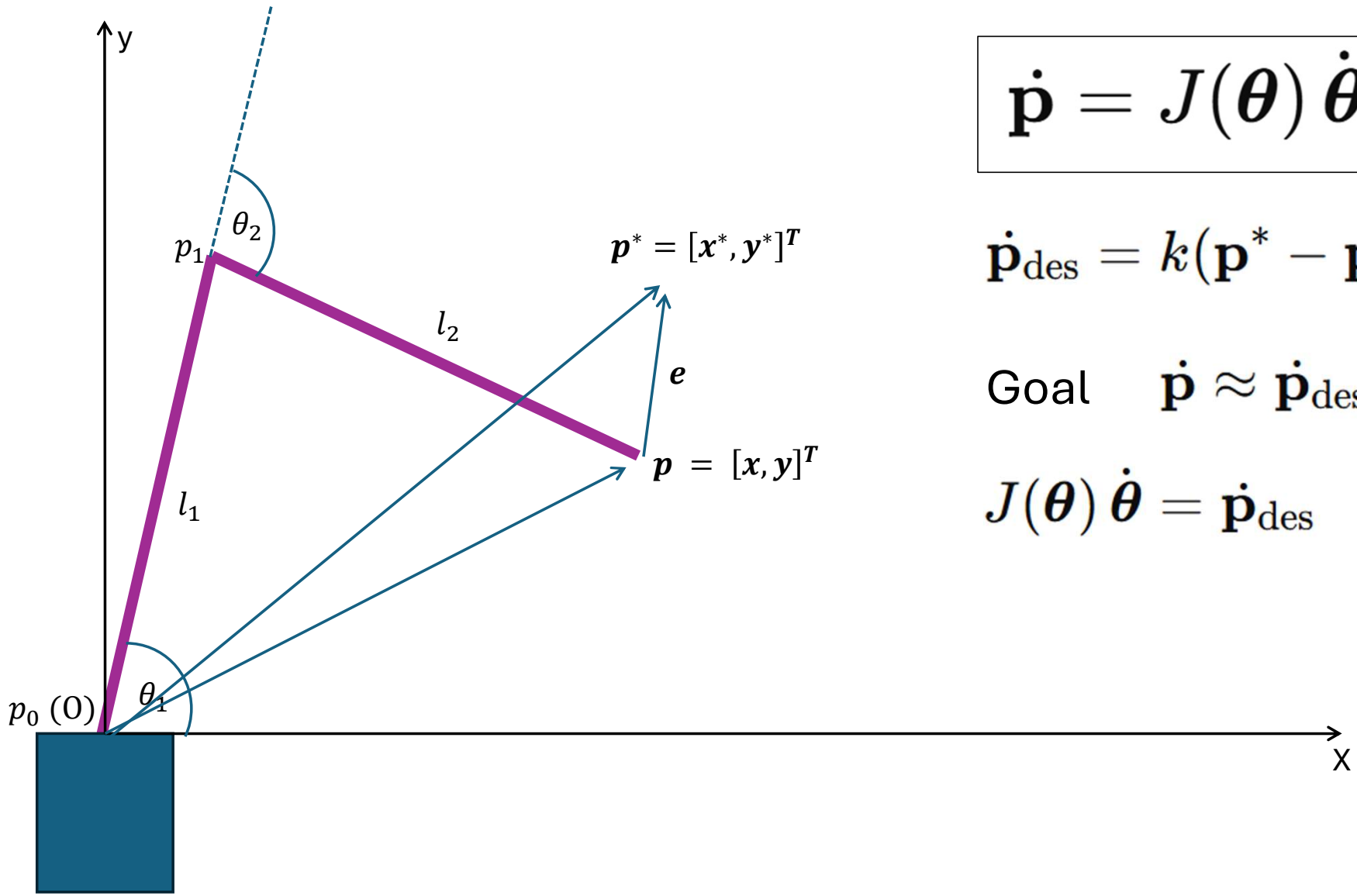
$$\frac{\partial x}{\partial \theta_2} = -l_2 \sin(\theta_1 + \theta_2)$$

$$\frac{\partial y}{\partial \theta_1} = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2)$$

$$\frac{\partial y}{\partial \theta_2} = l_2 \cos(\theta_1 + \theta_2)$$

$$J(\theta_1, \theta_2) = \begin{bmatrix} -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2) & -l_2 \sin(\theta_1 + \theta_2) \\ l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) & l_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

$$\boxed{\dot{\mathbf{p}} = J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}}} \quad !$$

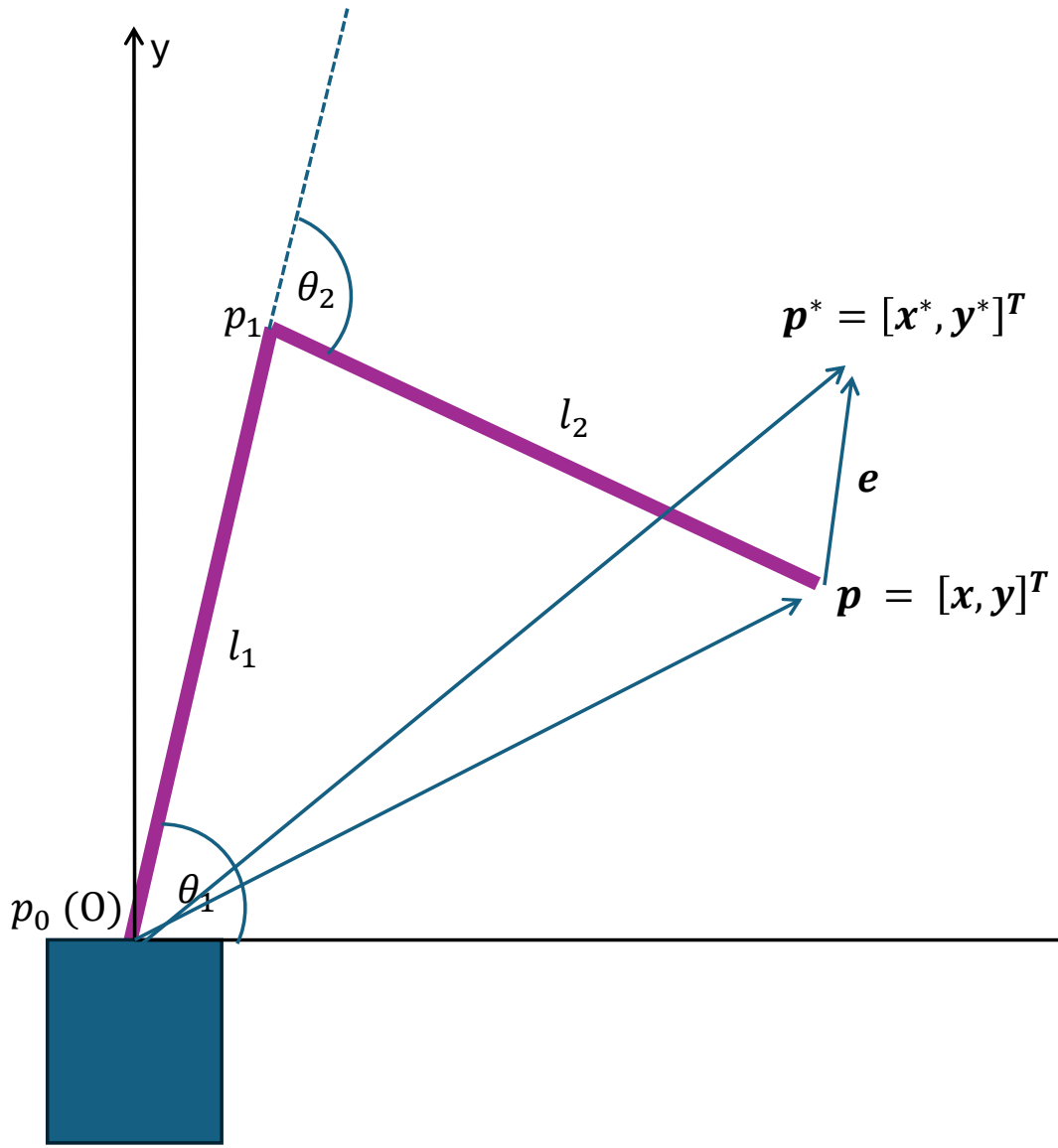


$$\boxed{\dot{\mathbf{p}} = J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}}}$$

$$\dot{\mathbf{p}}_{\text{des}} = k(\mathbf{p}^* - \mathbf{p})$$

Goal $\quad \dot{\mathbf{p}} \approx \dot{\mathbf{p}}_{\text{des}}$

$$J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}} = \dot{\mathbf{p}}_{\text{des}}$$



$$\boxed{\dot{\mathbf{p}} = J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}}}$$

$$\dot{\mathbf{p}}_{\text{des}} = k(\mathbf{p}^* - \mathbf{p})$$

Goal $\quad \dot{\mathbf{p}} \approx \dot{\mathbf{p}}_{\text{des}}$

$$J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}} = \dot{\mathbf{p}}_{\text{des}}$$

$$\boxed{\dot{\boldsymbol{\theta}} = J(\boldsymbol{\theta})^{-1}\,\dot{\mathbf{p}}_{\text{des}}}$$

!

$$J = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad \Rightarrow \qquad J^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$a = -l_1 \sin(\theta_1) - l_2 \sin(\theta_1 + \theta_2), \quad b = -l_2 \sin(\theta_1 + \theta_2)$$

$$c = l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2), \quad d = l_2 \cos(\theta_1 + \theta_2)$$

$$ad - bc = l_1 l_2 \sin(\theta_2)$$

$$l_1 = 1.0, \quad l_2 = 0.7, \quad \theta_1 = 30^\circ, \quad \theta_2 = 45^\circ$$

$$J(\theta_1, \theta_2) = \begin{bmatrix} -1.1761 & -0.6761 \\ 1.0472 & 0.1812 \end{bmatrix} \qquad \det(J) = 0.4950$$

$$J(\theta_1, \theta_2)^{-1} = \begin{bmatrix} 0.3660 & 1.3660 \\ -2.1157 & -2.3762 \end{bmatrix}$$

pick $\dot{\mathbf{p}}_{\text{des}} = \begin{bmatrix} 0.05 \\ 0.02 \end{bmatrix}$

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} = J^{-1} \dot{\mathbf{p}}_{\text{des}} = \begin{bmatrix} 0.0456 \\ -0.1533 \end{bmatrix} \text{ rad/s}$$

BUT WAIT!

- What if there IS NO inverse?

(e.g. 3 link arm in 2d!)

$$\dot{\mathbf{p}} \in \mathbb{R}^m, \quad \dot{\boldsymbol{\theta}} \in \mathbb{R}^n, \quad \boldsymbol{J} \in \mathbb{R}^{m \times n}$$

$$\boldsymbol{J} \in \mathbb{R}^{2 \times 3}$$

Moore–Penrose pseudoinverse ! J^+

$$\dot{\boldsymbol{\theta}} = J^+ \, \dot{\mathbf{p}}_{\text{des}}$$

$$J^+ = J^T \left(J J^T\right)^{-1} \qquad \dot{\mathbf{p}} \in \mathbb{R}^m, \quad \dot{\boldsymbol{\theta}} \in \mathbb{R}^n, \quad J \in \mathbb{R}^{m \times n}$$

Redundant robot $n > m$(more joints than task dimensions)

$$\boxed{\dot{\boldsymbol{\theta}} = J^T \left(J J^T\right)^{-1} \dot{\mathbf{p}}_{\text{des}}}$$

Moore–Penrose pseudoinverse ! J^+

$$\dot{\boldsymbol{\theta}} = J^+ \, \dot{\mathbf{p}}_{\text{des}}$$

$$J^+ = (J^T J)^{-1} J^T$$

$$\dot{\mathbf{p}} \in \mathbb{R}^m, \quad \dot{\boldsymbol{\theta}} \in \mathbb{R}^n, \quad J \in \mathbb{R}^{m \times n}$$

Underactuated $n < m$(fewer joints than task dimensions)

$$\boxed{\dot{\boldsymbol{\theta}} = (J^T J)^{-1} J^T \, \dot{\mathbf{p}}_{\text{des}}}$$

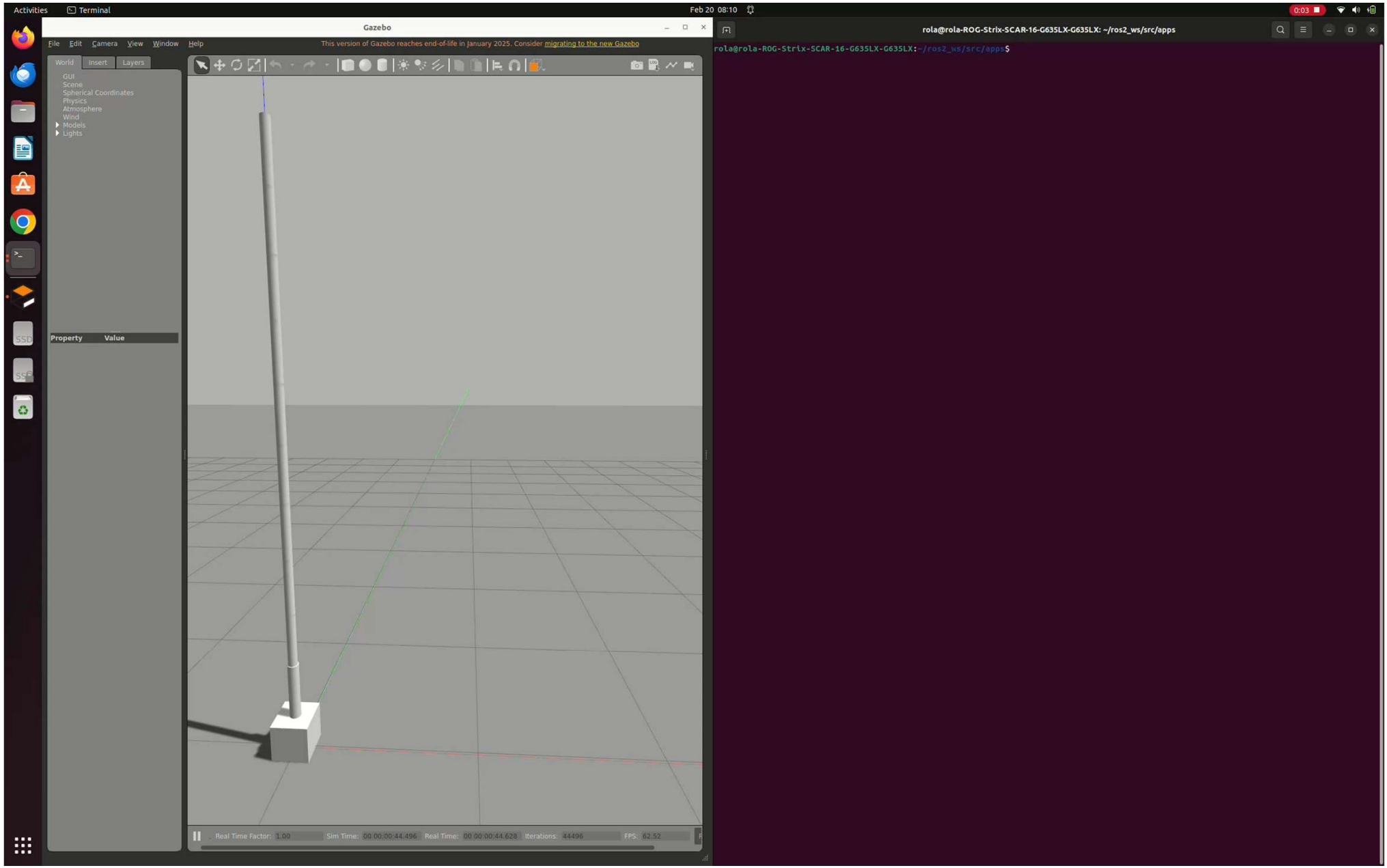
$$\dot{\boldsymbol{\theta}} = J^{+}\,\dot{\mathbf{p}}_{\text{des}} \qquad \text{where} \qquad J^{+} = \begin{cases} J^{-1} & \text{if } J \in \mathbb{R}^{n\times n} \text{ and invertible} \\ J^{T}(JJ^{T})^{-1} & \text{if } n > m \text{ and full row rank} \\ (J^{T}J)^{-1}J^{T} & \text{if } m > n \text{ and full column rank} \end{cases}$$

$$\det(J) \neq 0$$

Tasks:

- Implement IK for 2DOF manipulator (ros2 node; python)
- Implement Jacobian (Velocity kinematics) for 2DOF manipulator

- Extra: Implement Jacobian (Velocity kinematics) for 10DOF manipulator (ros2 node; python)


Activities
Terminal
Feb 20 08:10
Gazebo
File Edit Camera View Window Help
This version of Gazebo reaches end-of-life in January 2025. Consider migrating to the new Gazebo
World Insert Layers
GUI
Scene
Spherical Coordinates
Physics
Atmosphere
Wind
Models
Lights
Property Value
Real Time Factor: 1.00 Sim Time: 00 00:00:44.496 Real Time: 00 00:00:44.628 Iterations: 44496 FPS: 62.52
rola@rola-ROG-Strix-SCAR-16-G635LX-G635LX: ~/ros2_ws/src/apps
rola@rola-ROG-Strix-SCAR-16-G635LX-G635LX:~/ros2_ws/src/apps$